

EXPERIMENT: Prolog Program to Sum Integers from 1 to N

Aim

To write and implement a Prolog program to find the sum of integers from 1 to N using recursion.

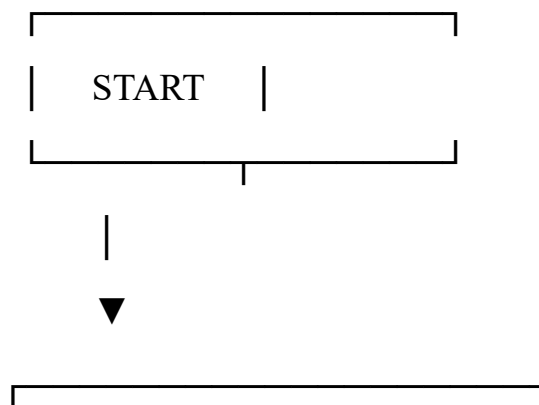
Objective

- To understand recursion in Prolog.
- To calculate the sum of integers from 1 to N.
- To understand facts, rules, and arithmetic operations in Prolog.
- To execute a simple mathematical problem using Prolog.

Algorithm

1. Start the program.
2. Read the value of N.
3. If $N = 0$, return 0.
4. Otherwise, calculate the sum of integers from 1 to $N-1$.
5. Add N to the calculated sum.
6. Repeat the process recursively until $N = 0$.
7. Display the final sum.
8. Stop.

Flowchart



| Read N |

|

|



| Is N = 0? |

|

| Yes | No



| Sum = 0 | | Sum(N-1) |

| + N |

|

|

|

|

|



|

|

|

| Repeat until |

|

| N = 0 |

|

|

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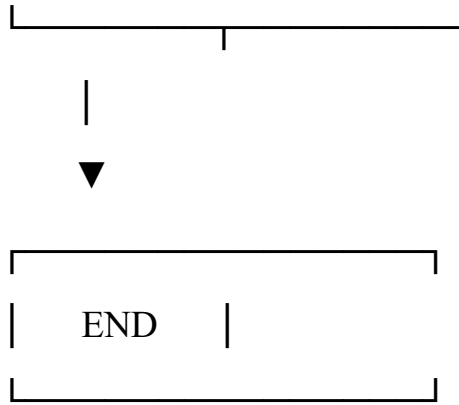
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| Display Sum |



CODE

```
def sum_recursive(n):  
    # Base case  
    if n == 0:  
        return 0  
    return n + sum_recursive(n - 1)  
  
def sum_iterative(n):  
    total = 0  
    for i in range(1, n + 1):  
        total = total + i  
    return total  
  
def sum_formula(n):  
    return n * (n + 1) // 2  
  
def display_steps(n):  
    print("\nStep-by-step calculation:")  
    total = 0
```

```

for i in range(1, n + 1):
    total = total + i
    if i == 1:
        print(i, end="")
    else:
        print(" +", i, end="")

print(" =", total)

def main():
    print("=" * 55)
    print("    SUM OF INTEGERS FROM 1 TO N")
    print("=" * 55)
    try:
        n = int(input("Enter the value of N: "))
    except ValueError:
        print("Invalid input! Please enter an integer.")
    if n < 0:
        print("Please enter a non-negative integer.")
        return
    print("\nN =", n)
    # Recursive method
    recursive_result = sum_recursive(n)
    iterative_result = sum_iterative(n)

```

```
print("\nResults")

print("-" * 55)

print("Using Recursion :", recursive_result)
print("Using Loop    :", iterative_result)
print("Using Formula  :", formula_result)

if n <= 20:

    display_steps(n)
else:

    print("\nN is large, so step-by-step output is not displayed.")

print("\nVerification")

print("-" * 55)

if recursive_result == iterative_result == formula_result:

    print("All three methods produced the same result.")
else:

    print("There is a difference between the results.")

print("\n" + "=" * 55)

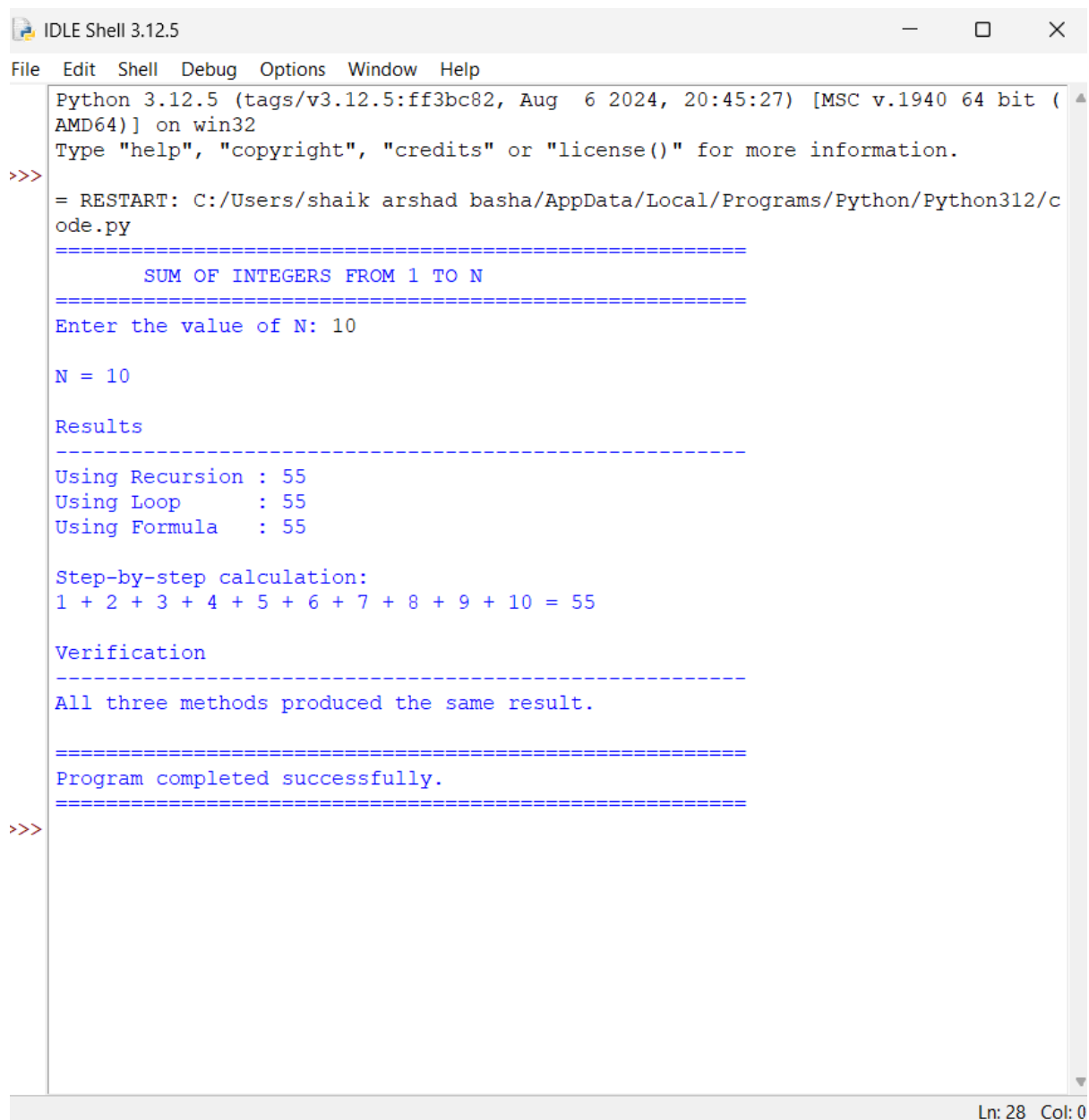
print("Program completed successfully.")

print("=" * 55)

if __name__ == "__main__":

    main()
```

Output:



```
IDLE Shell 3.12.5
File Edit Shell Debug Options Window Help
Python 3.12.5 (tags/v3.12.5:ff3bc82, Aug 6 2024, 20:45:27) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/shaik arshad basha/AppData/Local/Programs/Python/Python312/code.py
=====
SUM OF INTEGERS FROM 1 TO N
=====
Enter the value of N: 10

N = 10

Results
-----
Using Recursion : 55
Using Loop      : 55
Using Formula   : 55

Step-by-step calculation:
1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = 55

Verification
-----
All three methods produced the same result.

=====
Program completed successfully.
=====
>>>
```

Ln: 28 Col: 0

Result

The Prolog program was successfully implemented to calculate the sum of integers from 1 to N using recursion.

For example, when N = 10:

$$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = 55$$

Therefore, the output is 55.

Conclusion

The Prolog program successfully calculates the sum of integers from 1 to N. This program demonstrates the use of recursion, arithmetic expressions, rules, and base cases in Prolog.